

## Migration & Modernization

# Accelerating to the Cloud: How Stericycle rapidly migrated over 450 servers to Amazon EC2 and other AWS native services

by Sudhir Schar, Boatner Blankenstein, and Daniel Small | on 29 APR 2025 | in [Amazon EC2](#), [Best Practices](#), [Case Study](#), [Cloud Adoption](#), [Compute](#), [Customer Solutions](#), [Experience-Based Acceleration](#), [Foundational \(100\)](#), [Healthcare](#), [Intermediate \(200\)](#), [Migration Acceleration Program \(MAP\)](#), [Migration Portfolio Assessment](#) | [Permalink](#) | [Share](#)

*This post is co-written with Matthew Kozlowski from Stericycle, Inc.*

Discover how a five Person infrastructure team migrated 450+ servers for 24/7 operations, achieving cost savings and efficiency gains. This case study reveals their successful strategy. [Stericycle, Inc.](#) has been protecting what matters since 1989. Headquartered in Bannockburn, Illinois, this leader in compliance-based solutions serves healthcare organizations and businesses across North America. With over 13,500 employees and annual revenues of \$2.7 billion (FY2023), Stericycle provides essential services in regulated waste management, secure information destruction, and compliance solutions that protect people and brands while safeguarding the environment.

## The Challenge: Racing Against Time

In early 2024, Stericycle faced a critical deadline: their data center contract would expire on August 1, 2024, with an automatic one-year renewal if not terminated. While the team had initially planned a migration to [VMware Cloud on AWS](#), industry changes necessitated a rapid pivot to native Amazon EC2 services. This shift compressed their timelines dramatically, requiring the migration of:

- 450+ production servers
- Mission-critical healthcare applications
- Legacy systems dating back to 2003
- Complex networking dependencies
- Compliance-sensitive workloads

## The Solution

The AWS team developed a powerful three-pronged migration strategy that proved crucial to Stericycle's success. At its core, [AWS Application Migration Service](#) (MGN) automated server replication and minimized downtime, effectively handling both modern and legacy systems across their 450+ servers. MGN's ability to maintain application consistency while simultaneously migrating diverse workloads was essential for handling both modern and legacy systems across their 450+ servers.

The [Experience Based Acceleration](#) (EBA) methodology structured the migration into five manageable waves, each building on lessons from previous phases. This iterative approach allowed the team to validate each phase,

refine processes, and maintain continuous healthcare operations. By scheduling critical moves during maintenance windows and implementing robust testing protocols, the team achieved a smooth transition within their tight 60-day timeline.

The technical foundation proved equally crucial. A dedicated 10 Gbps circuit and AWS Direct Connect ensured consistent replication speeds and minimal latency, while separate replication subnets maintained data integrity. This infrastructure architecture delivered the high-throughput connectivity and security essential for healthcare operations, preventing service disruptions throughout the migration and maintaining compliance requirements.

## Key Solution Components

### Phased Migration Approach

- Application dependencies were carefully mapped to ensure related components were migrated together.
- Systematic server migration executed across five such groupings or waves.
- Strategic scheduling of critical system moves during Friday night maintenance windows to minimize business impact.
- Continuous validation and testing after each migration phase, ensuring systems were operational by Saturday morning.
- Iterative approach allowing for refinement of processes between waves.

### Technical Implementation

- AWS [Application Migration Service](#) (MGN) as the migration tool for most servers.
- Additional migration methods including VM Import/Export for sensitive legacy workloads.
- Implementation of a dedicated 10 Gbps circuit and separate replication subnet for migration traffic.
- Utilization of AWS [Direct Connect](#) for stable, high-throughput connectivity to AWS.
- Mix of automated replication processes and manual configurations for complex systems.
- Comprehensive pre and post-migration testing protocols to ensure migrated applications met or exceeded performance requirements.

### Architecture and Security

- Implementation of AWS [Well-Architected Framework](#) best practices.
- Design of robust security controls to meet healthcare industry compliance requirements.
- Development of a scalable infrastructure to support future growth and flexibility.
- Custom solutions to preserve critical networking configurations for key applications dependent on IP addresses.

Implementation of enhanced monitoring and alerting systems to ensure visibility during and after migration

This multi-faceted approach allowed Stericycle to maintain their aggressive timeline while ensuring the stability and security of their critical healthcare operations. The flexibility built into the migration strategy meant they could address the diverse needs of their complex environment efficiently, from modern applications to legacy systems dating back to 2003.

## Results Achieved

Stericycle's cloud migration delivered impressive results that went beyond meeting the deadline. The team successfully migrated their entire production environment ahead of schedule, completing the move of 450+ servers by late July 2024. During the process, they identified and eliminated unnecessary systems, reducing their infrastructure footprint by 15%.

The financial impact proved significant, with the move to AWS generating \$2.2 million in annual savings compared to their previous managed solution. These savings came primarily from eliminating managed service overhead costs while gaining the flexibility and scalability of cloud infrastructure.

Most importantly, the migration caused minimal disruption to Stericycle's critical healthcare operations. The careful planning and execution meant that end users and customers experienced virtually no impact during this major infrastructure transformation.

## Lessons Learned and Best Practices:

Stericycle's migration highlights key best practices: comprehensive dependency mapping, parallel environments for quick rollbacks, and robust testing protocols for each wave. Dedication specific maintenance windows for critical moves with next-day validation proved effective. A separate replication network ensured stable migration performance while maintaining normal operations. These strategies minimized disruptions and ensured a smooth transition to the cloud environment.

## Applying This Approach to Your Environment:

Start by mapping your application dependencies and establish parallel environments for safety. Schedule migrations during low-impact windows, implement thorough testing protocols, and set up dedicated network paths for replication. Scale this approach based on your organization's size and complexity, adjusting timelines accordingly.

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### About the Authors



#### Matthew Kozlowski

Matthew Kozlowski serves as Infrastructure Engineering Lead at Stericycle, where he specializes in managing the company's critical data center infrastructure. His expertise spans both on-

premises and cloud environments, with hands-on responsibility for server operating systems, backup solutions, storage systems, and AWS cloud deployments. Matt plays a key role in maintaining Stericycle's hybrid infrastructure, ensuring reliable operations across both physical data centers and cloud platforms.



## Sudhir Schar

Sudhir is a Sr. Customer Solutions Manager at AWS, focusing on Agriculture and Consulting Services customers. His work centers on accelerating customers' cloud transformation journeys through strategic guidance and hands-on enablement. Leading enterprise-wide initiatives as an EBA ambassador, he specializes in solution architecture, migration strategies, and transformation consulting. In his role, Sudhir drives customer success through structured approaches while ensuring architectural excellence and business value realization.



## Boatner Blankenstein

Boatner Blankenstein is a Senior Worldwide Go-To-Market Specialist for AWS, helping organizations navigate their journey from VMware environments to AWS. His expertise spans cloud architecture, security compliance, and enterprise infrastructure modernization. In his role, Boatner collaborates with global teams to share best practices and success stories, helping customers leverage AWS services to achieve their business objectives.



## Daniel Small

Daniel Small is a Senior Migrations Solutions Architect at Amazon Web Services (AWS). His role is to advise customers on migration best practices, strategies, and business transformation. He draws on years of experience running data centers, managing global networks, and securing data perimeters. Daniel enjoys finding solutions to complex, interdisciplinary challenges. He is a licensed skydiver and dabbles in glassblowing when not at a keyboard.